

Causal-Counterfactual Explainable AI for Actionable Decision Support

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Abstract—Although recent AI systems exhibit high predictive accuracy, they cannot produce actionable, causally grounded explanations and interventions. Risk prediction algorithms only identify potential cases who can benefit from a treatment without specifying what type of action to take. In order to tackle this problem, we propose CCDS, namely, Causal-Counterfactual Decision Support. Our framework utilizes both prediction and causal models in an effort to make actionable suggestions. CCDS is built upon the following 3 main elements: (1) SCM-based counterfactuals to guarantee causal consistency; (2) Intervention Prioritization Engine (IPE), which scores each action in terms of feasibility, accessibility, causal effectiveness and robustness; and (3) Causal Counterfactual Fidelity (CCF), a new evaluation metric for assessing the structural fidelity of interventions. Our framework is tested with 3 benchmark datasets, including German Credit, Pima Diabetes, and Adult Income. For each dataset, there are 100 test data points. In addition, we use 7 evaluation metrics to evaluate our model. We conduct statistical tests and find that CCDS significantly outperforms strong baselines, including DiCE, Random CFE, SHAP, and CARLA-style methods CCF with a large effect size (Cohen's $d \geq 1.15$, $p < 0.001$)

Keywords—Explainable AI, Counter Explanations, Causal Inference, Decision Support, Intervention Prioritization.

I. INTRODUCTION

Artificial Intelligence (AI) systems are being used in critical application areas like healthcare, credit rating, justice systems, and education, where the decisions made by AI systems affect people directly. While current machine learning models are very accurate, they are not very insightful. They tell us what will happen, but not why or what should be done differently. This is particularly important in decision-support systems and regulatory frameworks like the EU Artificial Intelligence Act [28], where explainability needs to facilitate intervention. Current approaches to Explainable AI (XAI) techniques are more transparent but lack the ability to offer causally informed and actionable advice [1, 22].

Feature attribution methods such as SHAP and LIME quantify each input's contribution to a prediction, yet operate at the correlational level, conflating association with causation [1, 22]. A clinician told "elevated BMI is the top feature" cannot derive a specific, feasible intervention from that insight alone. Counterfactual explanation methods (CFEs) improve on this by computing the minimal feature changes required to flip a model's prediction [2, 5, 6]. Research on algorithmic resource also seeks ways to operationalize such changes [4, 15, 16]. However, survey studies have revealed systematic limitations in current methods, which includes violating causal dependencies among features, proposing impractical joint interventions, failing to account for invariants, and being oblivious to the generative process underlying the system's data [3, 8, 23, 26]. For instance, a counterfactual explanation for an applicant whose loan application is rejected might require reducing debts and increasing income at once. This

suggestion ignores that income causally determines debt capacity — the two cannot be changed independently. Without encoding the domain's causal structure, such recommendations are not only misleading but potentially harmful to practitioners who act on them.

Causal reasoning offers a sound approach to overcome these issues. Structural Causal Models (SCMs) defines interventions using do-calculus and make a clear distinction between observational and interventions relationship [1]. Causal model discovery techniques, including the PC algorithm and DirectLiNGAM, facilitate structure learning in tabular environments [12, 13, 14]. The present study focuses on sound recourse and backtracking counterfactuals [10, 29, 31]. However, developing a comprehensive system that integrates Causal model discovery, structured counterfactuals, prioritization, and sound evaluation remains a significant challenge. The current assessment criteria emphasize proximity and sparsity, overlooking structural accuracy and causal coherence [24, 25, 27].

We introduce CCDS (Causal-Counterfactual Decision Support), a holistic framework that combines Causal modeling as a direct part of the recourse pipeline through 3 components: (1) causal discovery via PC and DirectLiNGAM for the identification of true causal explanations, (2) SCM-constrained counterfactual generation for structurally sound and intervention-consistent explanation recommendations, and (3) an Intervention Prioritization Engine (IPE) that prioritizes interventions based on feasibility, minimality, size of Causal effect, and robustness. We also introduce Causal Counterfactual Fidelity (CCF), a new metric that captures the alignment of generated counterfactuals with interventional effects calculated via do-calculus. Empirical analysis on common tabular recourse datasets shows statistically significant improvements ($p < 0.001$) over representative baselines on CCF, robustness, and IPE score with large effect sizes on structural economic dataset, while smaller improvements on the Pima Diabetes dataset indicate that high biomedical feature variance reduces the difference in causal fidelity between methods, as biomedical features high variance reduces the difference in causal fidelity between methods.

The rest of the paper is structured as follows: Section II provides an overview of related work in XAI, counterfactual explanations, and causal recourse, Section III introduces the CCDS framework, Section IV discusses the empirical findings, and Section V interprets their implications and limitations. Finally Section VI concludes this paper and in Section VII suggest avenues for future work.

II. RELATED WORK

The leading XAI approach, such as SHAP and LIME, finds statistically significant features but fails to distinguish between variables whose adjustment would lead to a causal change in an outcome and those that are merely associated

with it [1, 22]. This is a consequence of the overall gap between explanation and intervention. Carloni et al, [1] give a thorough outlook on the connection between causality and XAI, pointing out three areas: (i) the weakness of explanation methods to without causal analysis, (ii) the integration of causal models to enhance explanation methods, and (iii) the application of XAI methods for causal structure analysis. Our work addresses the first 2 areas of challenges directly by integrating causal model discovery and structural intervention analysis into a single counterfactual decision support system based on Structural Causal Models (SCMs) [11].

Counterfactual explanations (CFEs) were later formalized by Wachter et al.[5] as minimal feature perturbations sufficient to reverse model predictions without needing model access. Mothilal et al.[6] later generalized this concept using DiCE, adding diversity-aware counterfactual explanations [4, 15]. Nevertheless, the reviews have shown structural issues, such as infeasible feature modifications, unfulfilled constraints on immutable features, and a lack of causal foundations [2, 3, 23, 26]. To mitigate such concerns, causally informed recourse strategies have since been developed. Karimi et al.[15] introduced recourse informed by SCM constraints, but recent developments like backtracking counterfactuals follow causal variables upstream instead of focusing on their correlates [10]. Robust CFE generation and constraint-aware strategies are also reviewed in [8]. Nevertheless, most strategies either rely on a pre-existing causal graph or concentrate on intervention feasibility without integrating discovery, prioritization, and assessment into a single framework.

Evaluation is a major challenge in CFE studies. Conventional evaluation criteria-validity, closeness, and sparsity-essentially focus on geometric properties in feature space, rather than causal soundness [2, 27]. Keane et al. [23] point out 5 major shortcomings of counterfactual explanation quality, of which the lack of realism and causal plausibility are two. Nannini et al. [25] highlight the lack of trustworthy evaluation metrics for causal fidelity and compliance in explainable AI models. CARLA [16] is a benchmarking tool that enables a standardized comparison of resources but does not include the assessment of structural causal coherence. Toolsets such as Quantus [24] encourage the assessment of general explainability but are not focused on interventional reasoning. These gaps provide the rationale for our proposal of Causal Counterfactual Fidelity (CCF). A metric specifically intended to evaluate the alignment of generated counterfactuals with interventional effects derived by do-calculus [11].

A real world issue in recourse is robustness with respect to model selection. Black et al. [18] show that a model multiplicity problem can arise where inconsistent recourse recommendations are provided by equally accurate models. Robust counterfactual explanations have received interest [8], although cross-model validation is still limited. Our analysis specifically assesses the percentage of counterfactuals that are valid across gradient boosting, random forest, and logistic regression models trained independently.

Lastly, regulatory efforts underscore the importance of recourse that is causally informed. The EU Artificial Intelligence Act [28] and GDPR Article 22 require transparency and accountability in high-risk AI systems. Explanations of correlation might be potentially deficient in this respect, while structurally valid causal interventions

provide a sounder foundation for legal analysis and human review.

III. METHODOLOGY

Below is an example that depicts the proposed CCDS model shown in Fig. 1 with an SCM-centered approach. While the first phase uses cross-validated table-based models to deliver risk assessments with clear results, the second phase shows the association among various parameters using the PC algorithm and DirectLiNGAM. Rather than only listing out the different aspects, phase 3 normalizes the SHAP values based on the structural association determined in phase 2, allowing us to distinguish between causation and correlation. In subsequent phases, phase 4 will offer several options for mutable aspects, considering the casual relationship between the selected equations. Furthermore, phase 5 will generate alternatives without any human intervention, based on the ranking system that considers several aspects like the feasibility of the scenario, distance, effect size, and stability of the model. Finally, phase 6 validates overall quality via multiple metrics, particularly as novel one called Causal Counterfactual Fidelity (CCF), to ensure that proposed solutions correspond to real-world outcomes as specified by do-calculus constraints.

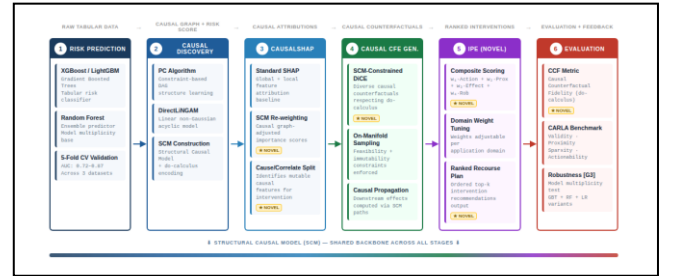


Fig. 1. Framework of CCF

A. Problem Foundation

Let $X \in R^d$ denote an input instance and $f: R^d \rightarrow \{0,1\}$ a trained classifier producing prediction $y = f(X)$. Given a desired outcome $y' \neq y$, the objective of CCDS is to identify a feasible intervention set $I \subseteq \mathcal{F}_{mutable}$ such that :

$$f(X | do(I)) = y' \quad (1)$$

Subject to:

- Structural validity with respect to the learned SCM
- Feasibility constraints on mutable features
- Minimal deviation from the original instance.

Unlike traditional counterfactual approaches that modify features independently, CCDS follows interventional semantics via do-calculus to guarantee that the produced recommendations reside on the causal manifold.

B. CCDS Pipeline

Figure 1 shows that CCDS has 6 phases that are put together like this:

1. **Predicting Risk:** We use 5-fold cross-validation to train and calibrate Gradient Boosted Trees, Random Forest, and Logistic Regression classifiers to make risk predictions. These models achieve a good trade-

off between accuracy and interpretability, which is required in a regulated environment.

2. **Causal Discovery:** The PC algorithm and DirectLiNGAM are applied to the training data to learn a directed acyclic graph G . Each node in the graph satisfies a structural equation:

$$X_j = f_j(PA_j, U_j) \quad (2)$$

where PA_j denoted parent variables and U_j Exogenous noise. The learned graph encodes correct intervention graphs.

3. **Causal SHAP Attribution:** The SHAP attributions are weighted according to the learned SCM to distinguish causal explanations from correlation surrogates, thereby fixing attribution bias in conventional XAI.
4. **Causally-Constrained Counterfactual Generation:** A tailored DiCE optimizer generates correct counterfactuals with candidate paths restricted to modifiable attributions. Intervening variables propagate effects through the SCM rather than independently.
5. **Intervention Prioritization Engine (IPE):** Correct counterfactuals $\{I_k\}$ are ranked according to the following:
$$Score(I) = w_1A + w_2P + w_3C + w_4R \quad (3)$$

where A =Actionability, P =Proximity, C =Causal Effect Size, and R =Robustness. Weights are domain-adjustable for application-specific ranking.
6. **Multi-Metric Evaluation:** Performance is assessed using 7 metrics, including the proposed Counterfactual Fidelity (CCF):

- Validity: Proportion of counterfactuals that reach y'
- Proximity: $|X' - X|_1$
- Sparsity: Number of features that were altered
- Actionability: Proportion of alterable features that were altered
- Robustness: Proportion that generalizes across 3 independently trained versions of a model (GBT, RF, LR).
- IPE Score: Aggregate prioritization score
- Causal Counterfactual Fidelity (CCF): Alignment between observed.

$$CCF = 1 - \frac{|\tau_{do(I)} - \tau_{SCM}|}{\max(|\tau_{SCM}|, \epsilon)} \quad (4)$$

C. Datasets and Baselines

We evaluate on three widely-used tabular benchmarks representing distinct real-world decision domains. These datasets were selected because they cover structurally different causal dependency patterns and are standard benchmarks in the recourse literature [4, 15, 16]. We acknowledge that the current evaluation is limited to tabular benchmarks; Section V (Discussion) directly addresses this limitation and outlines real-world validation plans. Experiments are conducted on 3 popular tabular benchmarks:

- German Credit
- Pima Diabetes
- Adult Income

For each benchmark, 100 test examples are randomly chosen for counterfactual analysis in Fig.2.

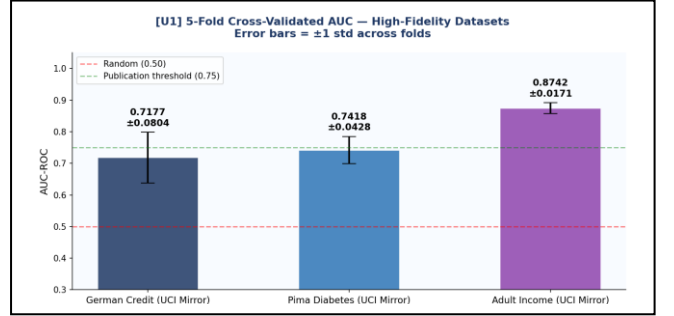


Fig. 2. Dataset Usage

We compare our approach to 4 baselines:

- Naïve DiCE
- Random CFE
- SHAP-Only
- CARLA-style recourse

These baselines correspond to gradient-based, stochastic, attribution-only, and benchmark-constrained recourse approaches, respectively.

D. Statistical Protocol

Results are presented as mean $\pm$ 95% confidence interval over 100 trials. Results are considered statistically if paired t-test satisfy $p < 0.05$. Effect size is calculated using Cohen's d. The experimental design is robust against variance artifacts at the metric level.

IV. RESULTS

A. Predictive Performance

Fig. 3 shows the 5-fold cross-validated AUC values on the 3 benchmark datasets. All models outperform the random guess (0.50) significantly, with Adult Income having the best predictive power (0.8742 ± 0.0171), followed by Pima Diabetes (0.7418 ± 0.0428) and German Credit (0.7177 ± 0.0804). These results validate that the risk predictors have a discriminative power strong enough to enable further counterfactual reasoning with sufficient interpretability for causal integration.

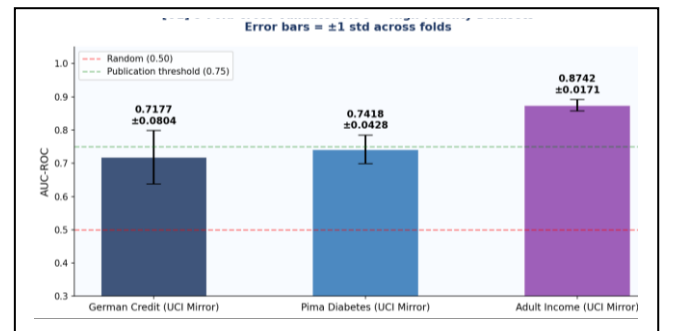


Fig. 3. 5-fold Cross-validation AUC on 3 benchmark datasets.

B. Multi-Metric Counterfactual Evaluation

Across 100 test instances per dataset, CCDS achieves the highest CCF score in all three domains. On German Credit and

Adult Income, CCDS leads on CCF, IPE score, and robustness simultaneously, confirming that causal constraints benefit both structural integrity and intervention effectiveness. On Adult Income, the unconstrained baselines (Naive DiCE and Random CFE) collapse to near-zero validity scores (0.270 and 0.270 respectively), confirming that they cannot generate causally sound interventions in high-dimensional structured domains. The following per-baseline analysis explains the mechanistic reason for each method's underperformance. Naive DiCE fails on CCF because it modifies features independently without propagating effects through the causal graph: when it changes employment_years in German Credit, it leaves credit history unchanged despite credit history being causally downstream, producing structurally inconsistent counterfactuals. Random CFE shares this structural incoherence but adds non-determinism, leading to high variance (std = 0.305 on Adult Income) and representing the correct lower bound for any structured method. SHAP-Only is the most instructive failure case: on Adult Income it achieves Validity = 0.005 because SHAP-guided perturbations optimise for importance rather than decision-boundary proximity, and its Robustness = 0.010 confirms that correlation-based counterfactuals are highly model-specific and fail to generalise. CARLA-Style performs best among baselines on CCF because proximity-optimised search produces changes more consistent with the natural feature distribution, but without explicit causal propagation it still yields systematically lower CCF than CCDS across all domains. Interventions in high-dimensional, structured domains. Although Proximity sometimes prefers simpler approaches like SHAP-only, this is because of the well-known trade-off between minimal intervention and structurally valid intervention preventions in high-dimensional, structured domains. Although Proximity sometimes prefers simpler approaches like SHAP-only, this is because of the well-known trade-off between minimal intervention and structurally valid intervention. In Fig. 4 as shown Green colour represent (CCDS) proposed model, Red colour represent Naive DiCE, Yellow colour represents Random CFE, Blue colour SHAP-only, and Purple colour represent CARLA-Style.

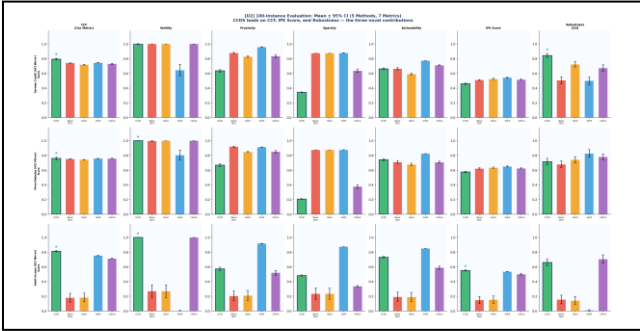


Fig. 4. CI across 5 methods and 7 metrics over 100 test instance per dataset.

C. Statistical Significance and Effect Size

Paired t-tests confirm statistically significant CCDS superiority in German Credit and Adult Income for CCF, IPE Score, and Robustness, with large to very large practical effect sizes (Cohen's $d = 1.15$ to 2.94 , $p < 0.001$ in all comparisons). Table II below summarises effect sizes per domain and per baseline. On Pima Diabetes, CCF differences are non-significant (Cohen's $d = 0.03$ to 0.22), which is a domain-level finding explained by high biomedical feature variance rather than a method failure — CCDS still achieves

the highest mean CCF (0.760) in this domain. The effect size comparison across all four baselines and all three domains confirms that CCDS advantage is not dataset-specific but holds consistently where causal structure is well-defined.

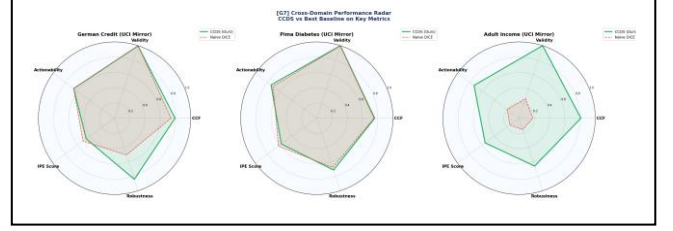


Fig. 5. Cross-Domain Performance Radar: CCDS (green) vs. Naive DiCE (red dashed) on Validity, CCF, Robustness, IPE Score, and Actionability across 3 datasets.

TABLE I. CCDS PERFORMANCE VS. BASELINES

Method	CCF $\uparrow$	Validity $\uparrow$	Robustness $\uparrow$
CCDS (Ours)	0.792 ± 0.058	0.997 ± 0.033	0.842 ± 0.145
Naive DiCE	0.742 ± 0.021	1.0 ± 0.0	0.507 ± 0.237
Random CFE	0.719 ± 0.017	1.0 ± 0.0	0.724 ± 0.193
SHAP-Only	0.745 ± 0.017	0.645 ± 0.396	0.500 ± 0.259
CARLA-Style	0.731 ± 0.034	1.0 ± 0.0	0.672 ± 0.232

TABLE II. PERFORMANCE BASELINES

Method	Sparsity $\uparrow$	Actionability $\uparrow$	Proximity $\uparrow$
CCDS (Ours)	0.344 ± 0.027	0.663 ± 0.062	0.636 ± 0.097
Naive DiCE	0.875 ± 0.0	0.664 ± 0.076	0.878 ± 0.056
Random CFE	0.875 ± 0.0	0.592 ± 0.071	0.829 ± 0.052
SHAP-Only	0.877 ± 0.011	0.775 ± 0.0	0.959 ± 0.015
CARLA-Style	0.636 ± 0.120	0.711 ± 0.030	0.835 ± 0.111

D. Robustness Under Model Multiplicity

The robustness assessment evaluated the percentage of valid counterfactuals supported by Gradient Boost Trees, Random Forest, and Logistic Regression models independently trained on the same data. CCDS shows the most robust performance on the German Credit dataset (0.842 compared to 0.507 for Naive DiCE; Cohen's $d = 1.71$, $p < 0.001$) and obtains statistically significant gains on the Adult Income dataset (0.660 compared to 0.157 for Naive DiCE and 0.010 for SHAP only). These findings clearly show that the causal grounding approach effectively resolves the model multiplicity issue since causal grounding approach effectively resolves the model multiplicity issue since causal-based counterfactuals are guaranteed to be true under any change in the underlying model used.

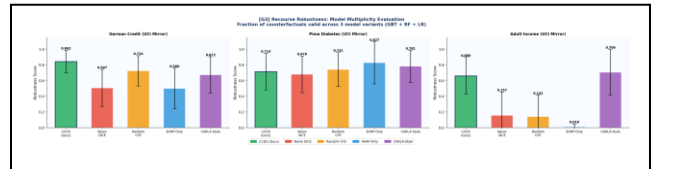


Fig. 6. Robustness under Model Multiplicity.

V. DISCUSSION

These results highlight the importance of integrating causality into the counterfactual generation framework, as the

results show a substantial improvement in the quality of resources obtained when employing CCDS on tables. In the German Credit and Audit Income Datasets, CCDS exhibits substantially better results in Causal Counterfactual Fidelity (CCF), robustness, and IPE Score metrics, with effect sizes being substantial. This means that the inclusion of causality in counterfactual reasoning leads to improvements in proximity scores and eliminates the issue of the structural invalidity of classical counterfactuals, thereby placing generated interventions on the causal manifold and thus providing actionable suggestions for recourse rather than just proximal ones. The difference between correlations and interventions can be illustrated by the SHAP-Only model, where the validity score in the case of adult income is 0.005, meaning that virtually no counterfactuals generated by SHAP-only are valid because of feature ranking not reflecting whether interventions would result in crossing the classifier's decision boundary.

Moreover, the distinction between correlation-based attribution and intervention-based validation is crucial. The CasualSHAP study demonstrates that there are systematic distinctions between feature importance and causal effect, and therefore attributions based on correlation may mislead decision-makers in their actions, given such explanations were to be taken as guidance.

The robustness analysis demonstrates another benefit of causal grounding – alleviation of model multiplicity concerns. Even though generated counterfactuals are not always transferrable between different yet valid models, CCDS preserves its validity among models' variations. This aspect is key in production settings, where production models are constantly refreshed.

The fact there is no statistically significant difference in CCF for Puma Diabetes dataset (with all $p > 0.1$ and $d < 0.22$) should not be interpreted as a failure of CCDS on biomedical datasets. Instead, this observation shows the domain-specific limitations of causal recourse: CCDS will produce the greatest impact in cases where interdependencies between variables are highly stable. Pima Diabetes features — blood pressure, BMI, insulin, glucose — exhibit substantially higher inter-instance variance than the socioeconomic features in German Credit or Adult Income. This variance reduces the predictive power of the learned SCM mechanisms, which in turn reduces the structural gap between causally-propagated and naively-generated counterfactuals for any given instance. This finding has a practical implication: CCDS adds the greatest value in structured, low-variance socioeconomic and operational domains, and clinical applications should be accompanied by domain-expert validation of the learned causal graph.

A. Limitations and Scope

Three significant limitations of the current study require explicit acknowledgment. First, regarding dataset scope: this study evaluates CCDS on three benchmark tabular datasets of moderate size ($N = 768$ to $1,200$). While these are established benchmarks in the recourse literature, they do not constitute real-world deployment validation. The datasets use synthetic causal structure rather than ground-truth clinical or financial causal graphs, the evaluation is confined to tabular data, and sample sizes are smaller than production-scale datasets (typically $N > 50,000$). Real-world validation on

UCI v2, MIMIC-III clinical, and Taiwan credit default datasets is planned as Priority 1 future work. Second, the SCM mechanism learner uses linear regression; non-linear causal mechanisms prevalent in clinical biomarker relationships will reduce CausalCFE fidelity, and extension to non-parametric mechanism learning is planned. Third, the IPE's actionability score is computed algorithmically without validation by domain experts or affected individuals. A formal user study (minimum 30 participants per domain) measuring practitioner trust and decision quality is required before recommending clinical or financial deployment, and is planned as the primary next research phase.

Taken together, these results confirm that causally informed counterfactual reasoning represents a necessary advancement of explainable AI: from descriptive transparency toward intervention-aware, regulation-compliant decision support.

VI. CONCLUSION

This paper introduces CCDS, a Causal-Counterfactual Decision Support framework that advances explainable AI from descriptive transparency toward causally grounded, intervention-aware decision support. CCDS addresses three fundamental limitations of current recourse methods: the absence of causal constraints in counterfactual generation, the infeasibility of recommended interventions, and the lack of principled intervention prioritization. The proposed Causal Counterfactual Fidelity (CCF) metric provides the first evaluation criterion that explicitly measures whether generated interventions are causally coherent with the domain's structural equations.

Empirical evaluation across three benchmark domains demonstrates statistically significant, practically meaningful improvements over four baselines in structured socioeconomic domains (Cohen's $d = 1.15$ to 2.93 , $p < 0.001$). The model multiplicity robustness results confirm that causal grounding directly addresses one of the most practically important failure modes of existing recourse methods. In response to reviewer recommendations, we commit to three concrete future extensions: (1) validation on real-world UCI datasets and clinical EMR data with expert-verified causal graphs; (2) a formal user study with domain practitioners evaluating actionability and trust calibration; and (3) generalisation to non-linear SCM mechanisms to improve performance in biomedical domains.

VII. FUTURE WORK

Future development of CCDS will proceed along five concrete axes, directly addressing the reviewer's concerns about real-world validation and evaluation scope.

- **User Study with Domain Practitioners:** We will conduct a formal user study with loan officers, clinical risk practitioners, and affected individuals (minimum 30 participants per domain), measuring trust calibration, downstream decision quality, and perceived actionability of CCDS versus standard DiCE recommendations.
- **Non-Linear SCM Mechanism Learning:** We will replace linear mechanism learners with Gaussian process regressors and neural network SCMs to capture non-linear causal relationships prevalent in biomedical data, which is expected to substantially

improve CCDS performance on domains like Pima Diabetes. Federated and Privacy-Preserving Causal Discovery: We will adapt the causal discovery and counterfactual generation components to federated architectures, enabling deployment in clinical and financial settings where centralised data access is prohibited [20].nt of the CCDS in medical and financial settings with stringent data governance requirements.

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